

BENEFITS

Why DI Is Good For Applications

THE PAYOFF FOR THE EXTRA INDIRECTION

1

Testability

Swap StripeGateway for a FakePaymentGateway in tests. No real API calls, no real charges, fast and deterministic tests of placeOrder() alone.

2

Loose Coupling

OrderService only knows about the PaymentGateway interface, never a concrete class — the two can change independently of each other.

3

Flexibility & Swappability

Switching from Stripe to PayPal is a one-line change where OrderService is wired up, not a rewrite of OrderService itself.

4

Follows the Dependency Inversion Principle

The "D" in SOLID: both high-level and low-level modules depend on abstractions, not on each other directly.

5

Centralized Configuration

Every "which implementation goes where" decision lives in one place — a composition root or container config — not scattered through the codebase.